

SIMATS ENGINEERING

ASSESSMENT TOOL 1

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COURSE NAME: CSA16Data Warehousing and Data Mining **COURSE CODE:** CSA1603

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Annexure A

Data Interpretation

Criteria	Understanding of Data (2.5)	Analysis (2.5)	Application of Concepts (2)	Conclusion (1.5)	Clarity (1)	Total(10)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I Siva Dakshitha K / 192425395 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Siva Dakshitha K

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID Age Monthly Medical Expense (₹) Health Risk Score

P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K SSE

1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q1 Data Set : Hospital Patient Segmentation

<u>Patient ID</u>	<u>Age</u>	<u>Monthly Medical Expenses (₹)</u>	<u>Health Risk Score</u>
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE values

<u>K</u>	<u>SSE</u>
1	600
2	350
3	220
4	200
5	190

Question:

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the following elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

SSE Means $\rightarrow$ Sum of Squared Errors.

We are using Elbow method

The reduction in SSE in

K	SSE	Reduction from Previous K
1	600	—
2	350	250
3	220	130
4	200	20
5	190	10

We observe that When

$K=2$ & 3 the SSE value is been reduced much and when K value is 4 & 5 the decrease is very small. This variation happens after the value of K is 3 . Therefore Elbow method suggest the value of K is 3

$\therefore$ Based on health risk Score 3 clusters have formed.

- i) Low risk - with low expense
- ii) Moderate risk - with moderate expense
- iii) High risk - with high expense.

Impact on Decision making:

$\Rightarrow$ For low risk group:

- i) preventive care
- ii) Routine monitoring

$\Rightarrow$ For moderate risk:

- i) Regular follow-up
- ii) Targeted intervention

$\Rightarrow$ For High risk:

- i) Immediate intervention
- ii) Intensive monitoring (close supervision)

OUTPUT:

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Clusterer

ChooseSimpleKMeans-init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

Use training set

Supplied test set

Percentage split

Classes to clusters evaluation

Store clusters for visualization

Set...

% 66

(Num) HealthRisk

Ignore attributes

StartStop

Result list (right-click for options)

11:41:37 - SimpleKMeans

11:44:04 - SimpleKMeans

Clusterer output

==== Run information ====

Scheme: weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Relation: hospital

Instances: 5

Attributes: 3

Age

MedicalExpense

HealthRisk

Test mode: evaluate on training data

==== Clustering model (full training set) ====

KMeans

Number of iterations: 3

Within cluster sum of squared errors: 0.047086194312283745

Initial starting points (random):

Cluster 0: 30,11000,70

Cluster 1: 55,25000,40

Cluster 2: 28,12000,65

Missing values globally replaced with mean/mode

Status

OK

Log

x 0

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Clusterer

ChooseSimpleKMeans-init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

Use training set

Supplied test set

Percentage split

Classes to clusters evaluation

Store clusters for visualization

Set...

% 66

(Num) HealthRisk

Ignore attributes

StartStop

Result list (right-click for options)

11:41:37 - SimpleKMeans

11:44:04 - SimpleKMeans

Clusterer output

Cluster 0: 30,11000,70

Cluster 1: 55,25000,40

Cluster 2: 28,12000,65

Missing values globally replaced with mean/mode

Final cluster centroids:

AttributeFull DataCluster#

(5.0)(1.0)(2.0)(2.0)

Age434257.529

MedicalExpense19200200002650011500

HealthRisk577537.567.5

Time taken to build model (full training data) : 0 seconds

==== Model and evaluation on training set ====

Clustered Instances

01 (20%)

12 (40%)

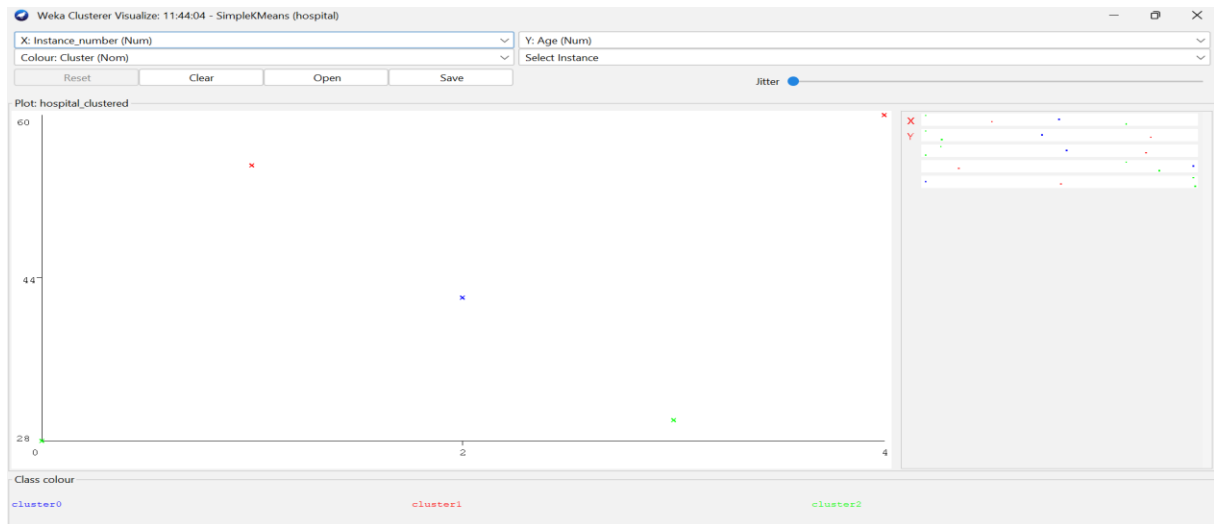
22 (40%)

Status

OK

Log

x 0



Q2. Dataset: E-Commerce Order Analysis

Order ID Product Category Order Value (₹)

O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

- Q2: Dataset : E-commerce order Analysis
- ② Dataset - A retailer applies hierarchical agglomerative clustering (HAC) to customer orders using appropriately encoded product category information and standardized order value. A dendrogram is generated. Interpret how clusters are progressively formed and determine the appropriate number of clusters by selecting a suitable cut-off level on the dendrogram. Explain how the selected clusters can help understand customer purchasing behaviour.

Order ID	Product Category	Order value (₹)
01	Electronics	55000
02	Clothing	25000
03	Electronics	57000
04	Furniture	30000
05	Clothing	27000

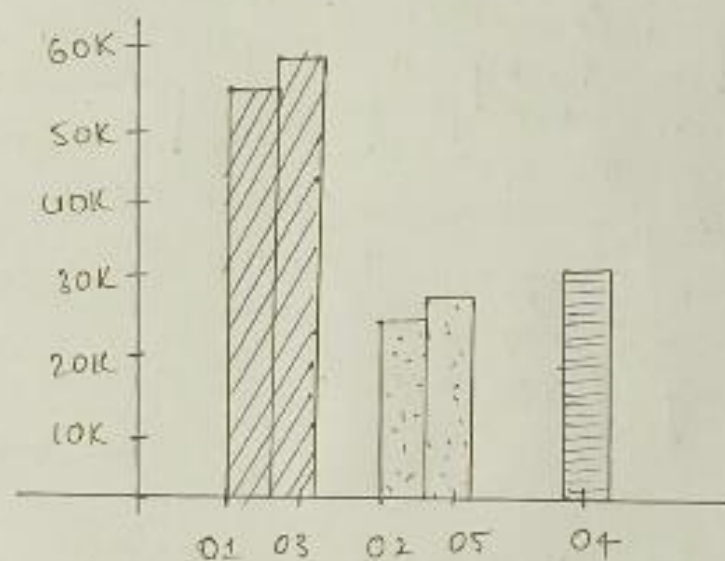
HAC

- It calculates the distance between the clusters
- Merges the two closest clusters
- Recalculates the distance
- Continue until it will become as one cluster

→ 01 & 03 forms 1st cluster - Electronics

→ 02 & 05 forms 2nd cluster - Clothing

→ 04 forms 3rd cluster - Furniture



But the optimal no. of clusters can not be determine by the given data

Step:

1) One-hot encoding

order	Electronics	Clothing	Furniture
01	1	0	0
02	0	1	0
03	1	0	0
04	0	0	1
05	0	1	0

Step 2: Normalize Order Value

Use Min-max normalization

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Here: $X_{\min} = 25,000$

$X_{\max} = 57,000$

Therefore:

$$\begin{aligned} O1 : \quad & \frac{55000 - 25000}{57000 - 25000} \Rightarrow \frac{30000}{32000} \\ & \Rightarrow \underline{\underline{0.9375}} \end{aligned}$$

$$O2 : \quad \frac{25000 - 25000}{57000 - 25000} \Rightarrow \frac{0}{32000} \Rightarrow \underline{\underline{0}}$$

$$O3 : \quad \frac{57000 - 25000}{57000 - 25000} \Rightarrow \frac{32000}{32000} \Rightarrow \underline{\underline{1}}$$

$$O4 : \quad \frac{30000 - 25000}{57000 - 25000} \Rightarrow \frac{5000}{32000} \Rightarrow \underline{\underline{0.15625}}$$

$$\begin{aligned} O5 : \quad & \frac{27000 - 25000}{57000 - 25000} \Rightarrow \frac{2000}{32000} \\ & \Rightarrow \underline{\underline{0.0625}} \end{aligned}$$

Order	Electronics	Clothing	Furniture	Normalized Value
01	1	0	0	0.9375
02	0	1	0	0
03	1	0	0	1
04	0	0	1	0.15625
05	0	1	0	0.0625

Step 3: Calculate Euclidean Distances

For two observations:

$$d(P, Q) = \sqrt{\sum_i (P_i - Q_i)^2}$$

01 & 03

$$\begin{aligned}
 d(01, 03) &= \sqrt{(1-1)^2 + (0-0)^2 + (0-0)^2 + (0.9375-1)^2} \\
 &\Rightarrow \sqrt{0+0+0+(-0.0625)^2} \\
 &\Rightarrow \sqrt{0.00390625}
 \end{aligned}$$

$$d(01, 03) \Rightarrow 0.015625$$

(02 & 05)

$$0.0625$$

$$\begin{aligned}
 d(02, 05) &= |0 - 0.0625| \\
 &= 0.0625
 \end{aligned}$$

0.4

$$d(04) = 0.0625$$

-The elements within cluster are small so it will be in small cluster only

Step 4:

Using the same matrix notation we are going to find Distance matrix.

Sid	01	02	03	04	05
01	0	1.697	0.063	1.616	1.663
02	1.697	0	1.732	1.423	0.063
03	0.063	1.732	0	1.647	1.697
04	1.616	1.423	1.647	0	1.417
05	1.663	0.063	1.697	1.417	0

Step 5: Hierarchical Agglomerative clustering we use Single linkage, where:

$$D(A, B) = \min d(x, y)$$

between members of the two clusters

Merge 1

Smallest distance:

$$01 - 03 \Rightarrow 0.0625$$

By seeing smallest distance.

$$C_1 = 01 + 03 \quad ; \quad C_1 = \{01, 03\} \Rightarrow 0.0625$$

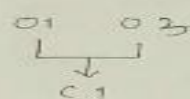
$$C_2 = 02 + 05 \quad ; \quad C_2 = \{02, 05\} \Rightarrow 0.0625$$

$$C_3 = 04 \quad ; \quad C_3 = \{04\} \Rightarrow 0.0625$$

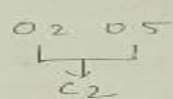
Merging of cluster can be done by finding the distance between $C_1 \& 04$, $C_2 \& 04$

Minimum of $C_1 \& 04$, $C_2 \& 04$

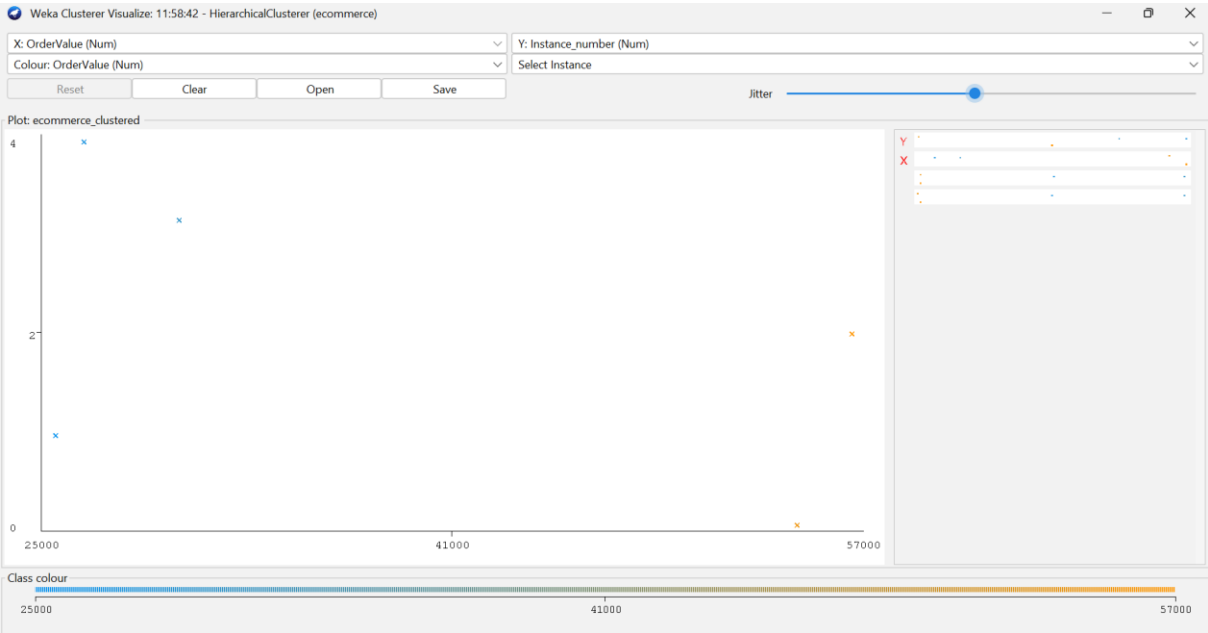
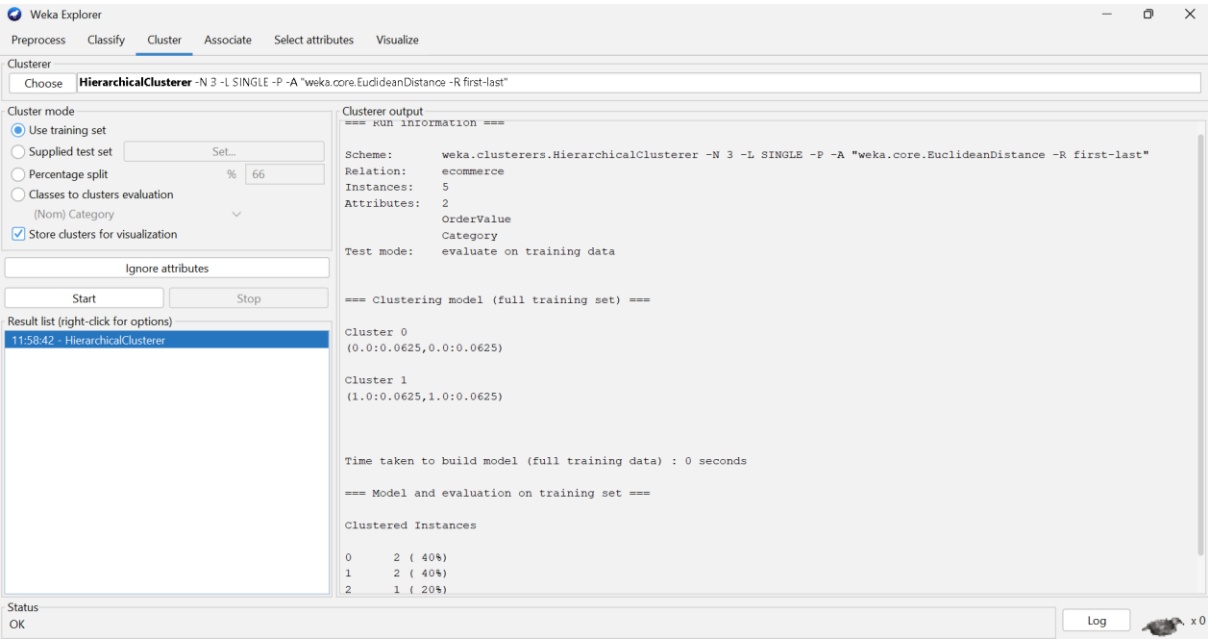
$$\begin{aligned} &C_1 \& 04 \\ &(01, 03) \& 04 \\ &\min((01, 04), (03, 04)) \\ &\min(1.616, 1.647) \\ &\underline{1.616} \end{aligned}$$



$$\begin{aligned} &C_2 \& 04 \\ &(02, 05) \& 04 \\ &\min((02, 04), (05, 04)) \\ &\min(1.423, 1.417) \\ &\underline{1.417} \end{aligned}$$



OUTPUT:



Q3. Dataset: Mobile App User Behaviour

User Daily Usage Time (min) Login Frequency In-App Purchase (₹)

U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Step 1: Identify the Features

we have 3 features for each user:

- Usage Time (minutes)
- Login Frequency
- Purchase Amount (₹)

Step 2: Observe the Data

Look at the user's behaviour

U1, U3, U5:

- High Usage
- High Login frequency
- Lower purchase amount

So they form one similar group

U2, U4:

- low Usage
- low login frequency
- High purchase amount

So they form another similar group

Step 3 compare the two groups

Features	G1: U1, U3, U5	G2: U2, U4
Usage	High	Low
Logins	High	Low
Purchase	Low	High

There is a clear difference between the two groups.

Step 4: Determine No. of Clusters

Since there are two distinct behavioral patterns
we choose: $K = 2$

Therefore:

No. of clusters = 2

Step 5: Name the clusters

cluster 1 - highly Engaged, lower-purchase Users

→ U1, U3, U5

cluster 2 - less engaged, high-Purchase Users

→ U2, U4

Step-6: Final Result

K = 2

cluster 1 = {U1, U3, U5}

cluster 2 = {U2, U4}

I analyzed three features - usage, log in, frequency, and purchase amount. The users show two clearly different behavioural patterns. Therefore, the appropriate number of clusters will be formed is "2".

OUTPUT:

The screenshot shows the Weka Explorer interface with the EM clustering algorithm applied to the 'mobile_users' dataset. The 'Clusterer' tab is active, and the 'Run' button has been clicked. The 'Clusterer output' pane displays the following information:

```
==== Run information ====
Scheme:      weka.clusterers.EM -I 100 -N 2 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 10
Relation:    mobile_users
Instances:    5
Attributes:   3
UsageTime
LoginFrequency
Purchase
Test mode:   evaluate on training data

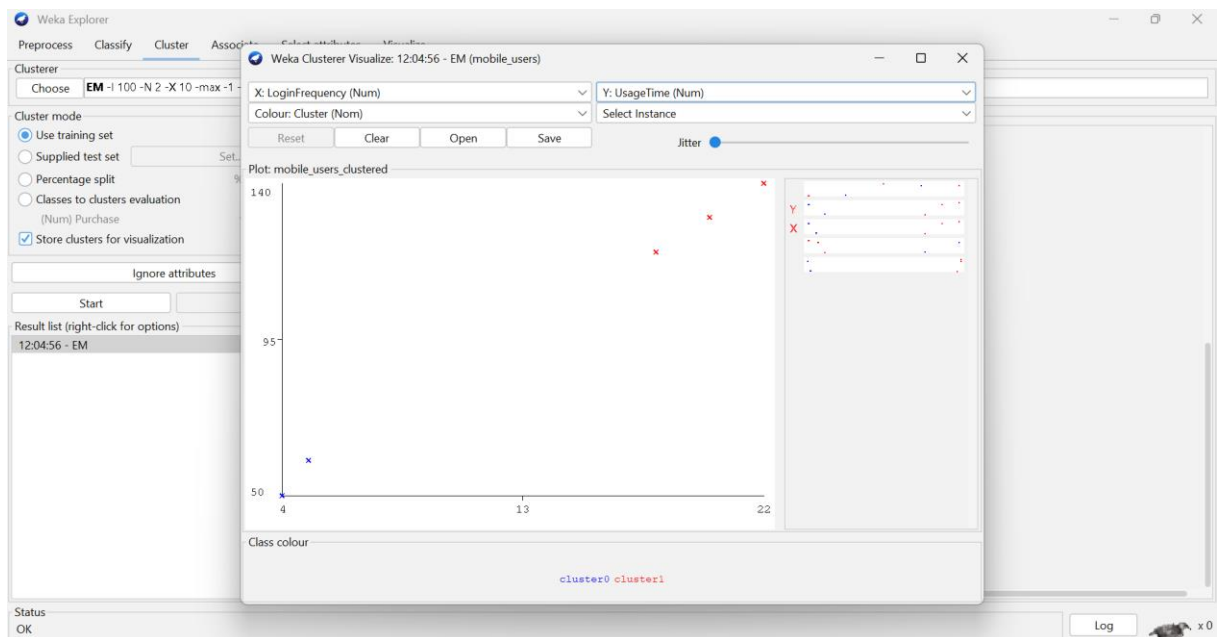
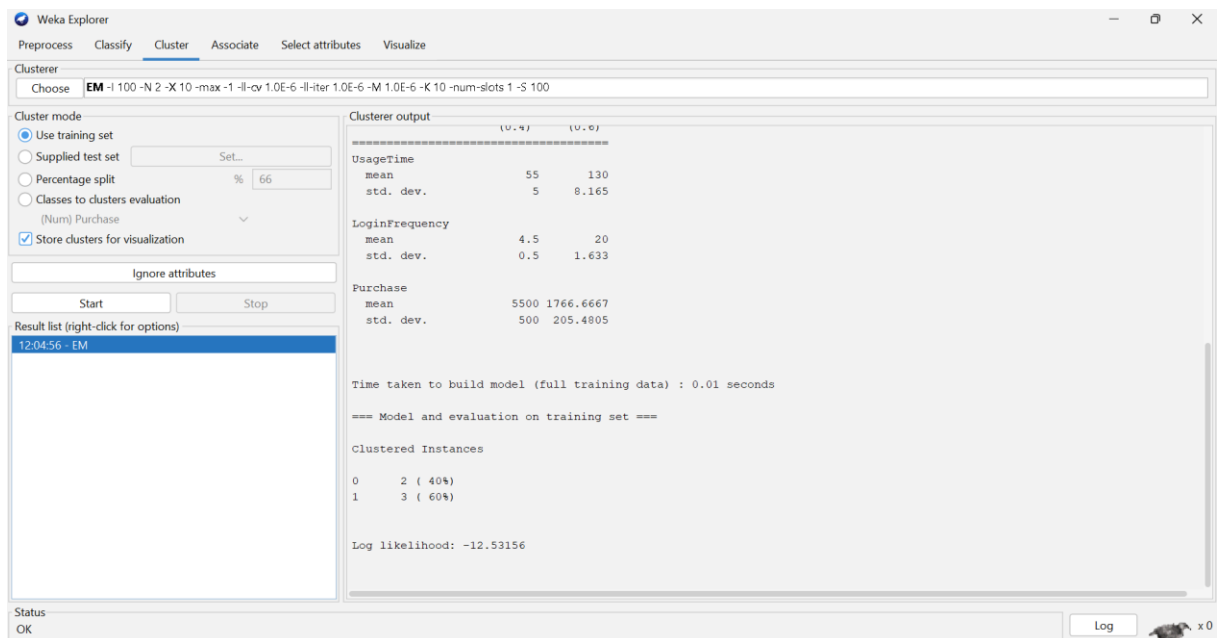
==== Clustering model (full training set) ====

EM

Number of clusters: 2
Number of iterations performed: 2

Attribute      Cluster      1
                (0.4)      (0.6)
-----
UsageTime
mean           55       130
```

The 'Status' bar at the bottom indicates 'OK'.



Q4. Dataset: Agricultural Crop Monitoring

Farm Rainfall (mm) Soil Fertility Score Crop Yield (tons)

F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Step 1: Calculate distances before normalization
We use Euclidean distance:

$$d(P, Q) = \sqrt{\sum_i (P_i - Q_i)^2}$$

for example,

$$\begin{aligned} d(F1, F4) &= \sqrt{(300-280)^2 + (8-9)^2 + (85-90)^2} \\ &= \sqrt{400 + 1 + 25} \\ &= \sqrt{426} \approx 20.64 \end{aligned}$$

Similarity the important distances are:

Pair	Distance
F1-F4	20.64
F2-F5	20.64
F1-F3	100.14
F2-F3	100.52
F3-F4	120.45
F3-F5	120.97
F3-F2	200.60
F4-F5	201.81

HAC before normalization

Using Single Linkage the smallest distance is:

$$F1 - F4 = 20.64$$

and

$$F2 - F5 = 20.64$$

Therefore,

$$C_1 = \{F1, F4\}$$

$$C_2 = \{F2, F5\}$$

$$C_3 = \{F3\}$$

Next, F3 Joins

C1, because:

$$\min[d(F3, F1), d(F3, F4)] = \min(100.14, 120.45) \\ = 100.14$$

So:

$$C1 = \{F1, F3, F4\}$$

Finally, F3 connects to C2 at 100.52, producing one cluster.

Step 2: Min-Max Normalization

The min-max formula is

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

• Rainfall: $x_{\min} = 280$, $x_{\max} = 520$

• Soil Fertility: $x_{\min} = 3$, $x_{\max} = 9$

• Crop Yield: $x_{\min} = 65$, $x_{\max} = 90$

Farm	Rainfall	Soil Fertility	Crop Yield
F1	0.0833	0.8333	0.8000
F2	0.9167	0.667	0.2000
F3	0.5000	0.5000	0.6000
F4	0.0000	1.0000	1.0000
F5	1.0000	0.0000	0.0000

Step 3: Euclidean Distance

Pair	Distance
F1-F4	0.2734
F2-F5	0.2734
F1-F3	0.5698
F2-F3	0.6669
F3-F4	0.8124
F3-F5	0.9274

the smallest distance are $F1-F4$ and $F2-F5$

Steps: HAC clustering

At 3 clusters:

$$C1 = \{F1, F4\}$$

$$C2 = \{F2, F5\}$$

$$C3 = \{F3\}$$

Thus similar form are grouped together

Steps: Interpretation

Before normalization rainfall dominates % because its numerical range is much larger.

After normalization, all features are scaled to 0-1, giving rainfall, Soil fertility, and crop yield a more balanced influence.

The final 3 clusters:

$$\{F1, F4\}, \{F2, F5\}, \{F3\}$$

OUTPUT:

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Cluster: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split % 66
- ☐ Classes to clusters evaluation (Num) CropYield
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

12:15:11 - SimpleKMeans

Cluster output

==== Run information ====

Scheme: weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Relation: agriculture

Instances: 5

Attributes: 3

Test mode: evaluate on training data

==== Clustering model (full training set) ====

kMeans

====

Number of iterations: 3

Within cluster sum of squared errors: 0.074722222222222

Initial starting points (random):

Cluster 0: 280,9,90

Cluster 1: 500,4,70

Cluster 2: 300,8,85

Missing values globally replaced with mean/mode

Status OK

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Cluster: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split % 66
- ☐ Classes to clusters evaluation (Num) CropYield
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

12:15:11 - SimpleKMeans

Cluster output

Cluster 0: 280,9,90

Cluster 1: 500,4,70

Cluster 2: 300,8,85

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data (5.0)	Cluster# 0 (2.0)	Cluster# 1 (2.0)	Cluster# 2 (1.0)
Rainfall	400	290	510	400
SoilFertility	6	8.5	3.5	6
CropYield	78	87.5	67.5	80

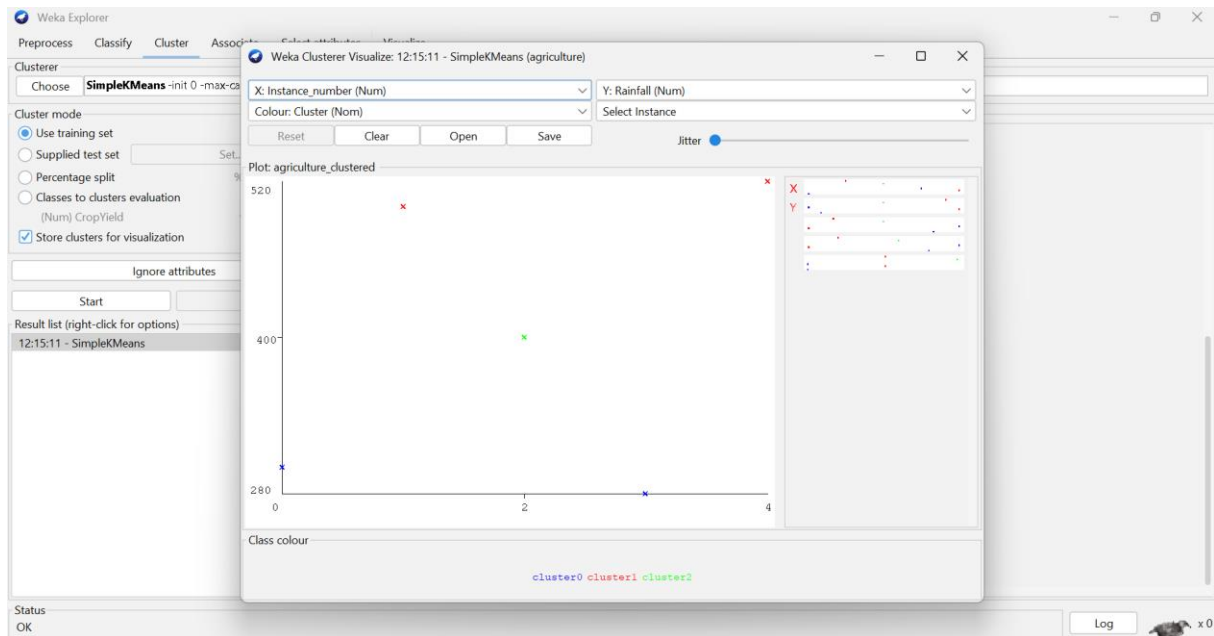
Time taken to build model (full training data) : 0.02 seconds

==== Model and evaluation on training set ====

Clustered Instances

Cluster	Count	Percentage
0	2	(40%)
1	2	(40%)
2	1	(20%)

Status OK



Q5. Dataset: Employee Performance Evaluation

Employee Productivity Score Attendance (%) Skill Assessment Score

E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model Score

K=2	0.48
K=3	0.68
K=4	0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

Given Silhouette Scores

<u>Model</u>	<u>Score</u>
$K=2$	0.48
$K=3$	0.68
$K=4$	0.52

Analysis

The Silhouette Score measures the quality of clustering. A higher score indicates better-defined and well-separated clusters.

- $K=2$: Score = 0.48 $\rightarrow$ Moderate clustering
- $K=3$: Score = 0.68 $\rightarrow$ Best clustering
- $K=4$: Score = 0.52 $\rightarrow$ Moderate clustering

Since $K=3$ has the highest Silhouette Score (0.68), it is the most suitable model.

Justification:

$K=3$ provides better similarity among employees within the same cluster and better separation between different clusters. It can effectively group employees based on their productivity, attendance and skill attainment.

conclusion:

Best Number of clusters = 3

The Selected K=3 model can help the company identify different employee performance groups, such as high, medium, and low performers. This supports better performance evaluation, training, recognition and employee development decisions.

OUTPUT:

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer
Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Skill
☒ Store clusters for visualization

Ignore attributes
Start Stop

Result list (right-click for options)
 12:15:11 - SimpleKMeans
 12:19:50 - SimpleKMeans

Cluster output

==== Run information ====

Scheme: weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -
 Relation: employees
 Instances: 5
 Attributes: 3
 Productivity
 Attendance
 Skill

Test mode: evaluate on training data

==== Clustering model (full training set) ====

KMeans
 =====

Number of iterations: 2
 Within cluster sum of squared errors: 0.27815523773183715

Initial starting points (random):

Cluster 0: 55,65,58
 Cluster 1: 62,75,68
 Cluster 2: 88,92,84

Missing values globally replaced with mean/mode

Status
OK

Log x0

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer
Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Skill
☒ Store clusters for visualization

Ignore attributes
Start Stop

Result list (right-click for options)
 12:15:11 - SimpleKMeans
 12:19:50 - SimpleKMeans

Cluster output

Cluster 0: 55,65,58
 Cluster 1: 62,75,68
 Cluster 2: 88,92,84

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data	Cluster# 0	Cluster# 1	Cluster# 2
	(5.0)	(1.0)	(1.0)	(3.0)
Productivity	75.2	55	62	86.3333
Attendance	83.2	65	75	92
Skill	74.4	58	68	82

Time taken to build model (full training data) : 0.01 seconds

==== Model and evaluation on training set ====

Clustered Instances

0	1 (20%)
1	1 (20%)
2	3 (60%)

Status
OK

Log x0

